



Spectrometer Calibration Certificate

Model: 1.129 S/N: 29G10004

Channels: 31 Size Channels (0.25µm – 32µm)

Firmware Version: 12.30

Revision: H

Calibration method:

The reference unit is calibrated with NIST certified PSL particles and the calibration is verified every six months. This is a worldwide accepted standard method referring to PTB Braunschweig and we therefore guarantee the traceability of our calibration. The absolute size calibration of the reference unit is transferred to the candidate unit with a calibration procedure using polydisperse dolomite particles.

Instruments used for Calibration

- Reference instrument model 1.109 S/N: 9G100016
- Oscilloscope Hameg HM507; S/N: 060210471
- Defender 520-M Revision C1 0.05 l/min. – 5 l/min.; S/N: 119942
- Calibration tower Model: 7851

Calibration Material:

- Reference unit: NIST certified monodisperse PSL particles with different diameters
- Candidate unit: Micro Dolomit DR90 polydisperse powder (0.30µm – 180µm)

Tolerance Ranges:

- Sample Flow Rate: 1.2 l/min ± 5%
- Count Correlation: ± 3% at 1µm
- Count Calibration: ± 3% at Concentrations ≥ 500 P/l
- Relative Mass Deviation: ± 3% **or** ± 2µg/m³ to Reference Unit

<u>Measurement values at calibration tower</u>			
Mean Value	Reference Unit	Calibrated Unit	Deviation
total dust	452,2 µg/m ³	451,6 µg/m ³	0,6 µg = 0.13 %
Sample volume: 0.018 m ³ /Sample time : 15 Min.			

<u>Measurement values at ambient air</u>			
Mean Value	Reference Unit	Calibrated Unit	Deviation
total dust	25,5 µg/m ³	27,1 µg/m ³	1,6 µg = 6,27 %
Sample volume: 0.0388 m ³ /Sample time : 32 Min.			

We hereby confirm that this instrument has been successfully calibrated and passed the mass test. All work has been done by qualified and trained staff of GRIMM Aerosol Technik.

This calibration is valid until 2015-02-28
Pouch, 2014-01-27


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